MÁXIMO CLEMENTE DESIDERIO RETAMAL ROJAS

Computer Science Engineer | Universidad de Chile

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PROFESSIONAL PROFILE

Computer Science Engineer from Universidad de Chile, specializing in software engineering, Data Science, Machine Learning, and distributed architectures. Proven experience building real-time data pipelines, integrating Affective Computing and Brain-Computer Interfaces (BCI) into 3D environments (Unity/Python), full-stack web development (Django/Vue.js), and cloud deployments (AWS, Docker). Main areas of interest include software engineering, cybersecurity, and human-computer interaction (HCI), with a passion for solving complex engineering challenges.

PROFESSIONAL EXPERIENCE & PROJECTS

Bachelor's Thesis / Capstone Engineering Project | **Universidad de Chile** *August 2025 – 2026 • Santiago, Chile*

- Designed and implemented a Python pipeline for real-time ingestion and filtering of electroencephalographic (EEG - Emotiv EPOC+) signals and processing of biometric datasets (DREAMER).
- Extracted frequency-domain features (Band Power, FFT, Butterworth IIR filters, DC removal) and trained ML models (SVM, Random Forest) to classify emotional states (Valence & Arousal).
- Developed temporal smoothing modules (circular buffers, EMA) and asynchronous streaming via TCP sockets to Unity 3D (C#), dynamically adapting the graphical environment (audio, lighting, and interactive guidance).

BCI & Machine Learning Research Project | **Universidad de Chile** *2023 – 2024 • Santiago, Chile*

- Developed a multi-threaded Python desktop application (Tkinter, Emotiv Cortex API) for real-time EEG telemetry acquisition (Emotiv EPOC+) and synchronized visual stimuli for mental command training.
- Designed experimental protocols and conducted empirical trials with 30 participants, capturing and structuring brainwave datasets across 7 distinct mental commands.
- Built non-linear dimensionality reduction pipelines using Autoencoders (Deep Learning) and PCA, evaluating clustering techniques to accelerate cross-user BCI transfer learning.

Software & Data Science Engineering Intern | **Perceptron SpA** *December 2022 – January 2023 • Santiago, Chile*

- Developed a Django/React web dashboard for production data management and predictive weight/slaughter models (Agrosuper), implementing asynchronous task queue tracking in backend and database.
- Built Python data transformation pipelines (hierarchical structures & dictionaries) in collaboration with mathematical engineers to feed linear optimization models for raw material recipes (**Cecinamix**).
- Refactored price elasticity calculation engines (**Revenew**) to compute store-branch-level metrics, and managed cloud deployments on AWS (EC2, Route 53), Docker, and Nginx.

Frontend Developer | **Fundación Arturo López Pérez (FALP)** *March 2022 – July 2022 • Santiago, Chile*

- Developed a responsive, component-based web frontend in Vue.js for an educational and community support platform for breast cancer patients.
- Implemented multimedia modules for streaming and publishing educational blogs, videos, and audio content, complete with an interactive commenting system.
- Built direct messaging interfaces and inquiry submission flows connecting patients with clinical case managers and medical staff for healthcare guidance.

Software Engineering Intern | **Wisetrack Corp** *January 2020 – March 2020 • Santiago, Chile*

- Translated SQL Server stored procedures to PostgreSQL for cleaning and structuring telemetry alerts in the **Third Eye** driver safety platform.
- Programmed a Python ETL daemon on a Linux VM to extract, transform, and replicate telemetry data into PostgreSQL and MySQL every 10 seconds.
- Stabilized and modularized facial recognition microservices (**MOBICUA**), integrated AWS Rekognition, evaluated IBM Watson AI services, and built analytics dashboards in Qlik Sense.

EDUCATION

Bachelor of Science & Engineering Degree in Computer Science (Highest Distinction) *2016 - 2026 • Santiago, Chile*
Universidad de Chile — Faculty of Physical and Mathematical Sciences (FCFM)

TECHNICAL SKILLS

Programming Languages:	Python, SQL, C#, JavaScript, C, Java, HTML/CSS.
ML & Data Science:	Pandas, NumPy, BCI/EEG, SVM, Random Forest, Autoencoders, PCA, Clustering.
Web & 3D Development:	Vue.js, Django, TCP Sockets, Unity 3D (C#), Unity 6, Tailwind CSS.
Databases:	PostgreSQL, MySQL, SQL Server, SQLite, MongoDB.
Cloud & DevOps:	AWS (EC2, Route 53), Docker, Nginx, Linux (Daemons, Shell), Windows, Raspberry Pi.
Languages:	Spanish (Native), English (Advanced).